RAG Assignment: Baseline vs. Re-Ranking

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Document: IFAB Laws of the Game 2026/27 (260 pages, ∼300 chunks)
Models: all-MiniLM-L6-v2 (bi-encoder), ms-marco-MiniLM-L-6-v2 (cross-encoder), Gemini 3.5 Flash-Lite (LLM)
Code: `https://github.com/neill-nil/LMA_RAG_assignment`

1 Architecture Comparison

A bi-encoder encodes the query and each chunk separately into fixed vectors and compares them with cosine similarity. Chunk vectors can be computed once and stored in an index, so search is very fast. But the query and chunk run independently where each chunk is squeezed into a single vector before the question is known, so fine details like negations ("may not"), exceptions, or which word a term attaches to get blurred. It mostly rewards overall topic and word overlap.

A cross-encoder takes the query and chunk together as one input, so self-attention runs across the tokens of both. Every query word can attend to every chunk word, so it can check whether the chunk actually answers the question rather than just being about the same topic. The main limitation is that nothing can be precomputed: every (query, chunk) pair needs a full forward pass at query time.

2 Qualitative and Rank Shift Analysis

Out of 8 test queries, the correct chunk was outside the baseline top 3 in 5 cases, and re-ranking moved it to rank 1 in all 5 (Table 1):

- **Keyword trap:** "Can a goal be scored directly from a kick-off?" Correct chunk (Law 8) went from rank 5 to rank 1. The baseline top 3 were all Goal Kick / Penalty Kick chunks, pulled in by the shared words "goal" and "kick".
- **Corner case:** ball going straight into the kicker's own goal from a kick-off. Rank 9 to rank 1.
- **Negation / role limit:** "Can a replaced goalkeeper take a kick in the penalty shoot-out?" Rank 6 to rank 1. The baseline confused "penalty shoot-out" (Law 10) with "penalty kick" (Law 14).
- **Buried chunk:** "When does the referee's authority to show cards begin and end?" Rank 8 to rank 1.
- **Negation:** "When is an offside player not penalised?" Rank 4 to rank 1.

Answer 1 vs. Answer 2. Because the prompt forces the LLM to answer only from the given context, the baseline did not hallucinate. In the first four cases it said "Not found in the provided context", while the re-ranked pipeline gave the correct rule with a citation. In the offside case, the baseline gave a partial answer: it listed the goal kick/throw-in/corner exceptions but left out the main clause that simply being in an offside position is not an offence. Answer 2 included it.

#	Type	Query	Rank	Answer 1 (baseline top 3)	Answer 2 (re-ranked top 3)
1	Keyword trap	Can a goal be scored directly from a kick-off?	5 → 1	Not found in the provided context.	Yes, a goal may be scored directly against the opponents from the kick-off [C1].
2	Corner case	What happens if the ball goes straight into the kicker's own goal from the kick-off?	9 → 1	Not found in the provided context.	If the ball directly enters the kicker's goal from the kick-off, a corner kick is awarded to the opponents [C1].
3	Negation / role limit	Can a replaced goalkeeper take a kick in the penalty shoot-out?	6 → 1	Not found in the provided context.	No, a replaced goalkeeper takes no further part and may not take a kick [C1].
4	Buried chunk	When does the referee's authority to show cards begin and end?	8 → 1	Not found in the provided context.	The referee has the authority to show yellow or red cards from entering the field of play at the start of the match until after the match has ended, which includes the half-time interval, extra time, and penalties (penalty shoot-out) [C1].
5	Negation	When is a player in an offside position not penalised for being offside?	4 → 1	There is no offside offence if a player receives the ball directly from a goal kick, a throw-in, or a corner kick [C1, C2].	A player in an offside position is not penalised if they receive the ball directly from a goal kick, a throw-in, or a corner kick [C3]. Additionally, simply being in an offside position is not an offence, and a player is only penalised upon becoming involved in active play at the moment the ball is played or touched by a team-mate [C1].
6	Exception (control)	What happens if a team has fewer than seven players during penalties?	2 → 1	The referee must not abandon the match if a team is reduced to fewer than seven players during penalties [C2].	The referee must not abandon the match if a team is reduced to fewer than seven players during penalties (penalty shoot-out) [C1].
7	New 2026/27 rule (control)	What happens if the goalkeeper holds the ball in their hands for too long?	2 → 1	If a goalkeeper controls the ball with their hand(s)/arm(s) inside their penalty area for more than eight seconds before releasing it, a corner kick is awarded to the opponents [C2].	If a goalkeeper inside their penalty area controls the ball with their hand(s)/arm(s) for more than eight seconds before releasing it, a corner kick is awarded to the opposition [C1, C2].
8	Document-specific (struck-out text)	Is the kicker cautioned if both the penalty taker and the goalkeeper commit an offence at the same time?	1 → 1	No, the kicker is not cautioned if both the penalty taker and the goalkeeper commit an offence at the same time, as the reference to the kicker being cautioned in this scenario has been removed [C1].	No, the kicker is not cautioned because the reference to the kicker being cautioned for offending at the same time as the goalkeeper was removed [C1].

Table 1: Failure queries. *Rank* is the position of the chunk containing the correct rule among the 10 candidates (bi-encoder → cross-encoder); the LLM only sees the top 3. Queries 1–5 are cases where re-ranking

Without the strict prompt, the LLM would likely have filled the gap from memory. When asked about the goalkeeper holding the ball too long with no context, Gemini gave the old rule (6 seconds, indirect free kick). With retrieval it gave the 2026/27 rule (8 seconds, corner kick).

The other three queries were controls where both pipelines already had the right chunk in the top 3. Re-ranking only moved it from rank 2 to 1, and both answers were correct.

Document-specific issue. The "Law changes" section shows deleted rules as struck-through text, and PDF extraction keeps that text as if it were current. For the question on whether the kicker is cautioned when kicker and goalkeeper offend together, both pipelines retrieved a chunk containing the deleted sentence "the kicker is cautioned". The LLM still answered correctly because the same chunk explained that the rule was removed. Re-ranking cannot fix this: it is a parsing problem.

3 Latency Profiling

Stage (CPU, 8 queries × 5 runs)	**Mean**	**Median**	**Max**
Pipeline 1: bi-encoder retrieval (top 10)	9.8 ms	9.4 ms	24.3 ms
Re-rank step only: cross-encoder (10 pairs)	267.5 ms	275.2 ms	344.0 ms
Pipeline 2: retrieval + re-rank	277.4 ms	283.6 ms	352.1 ms
LLM answer generation (Gemini, one live call)	∼1.0–1.2 s	–	–
Cross-encoder over all 301 chunks (one query)	8.2 s	–	–

Table 2: Runtime of the baseline retrieval vs. the re-ranking pipeline.

On CPU, bi-encoder retrieval of the top 10 took about 10 ms per query. Re-ranking 10 candidates with the cross-encoder took about 270 ms, so Pipeline 2 is roughly 28× slower at the retrieval stage (∼280 ms vs. ∼10 ms; Table 2). In absolute terms this is still small compared to the LLM call itself (∼1 s), so adding re-ranking only increases it by a very small amount without a significant difference.

4 Scaling Question

A cross-encoder has to run a full transformer pass for every (query, chunk) pair, and none of it can be cached because the query is part of the input. Cost therefore grows linearly with corpus size on every query. Even on this one 300-chunk document, scoring all chunks took about 8.2 s per query, compared to 0.27 s for 10 candidates. For a real collection with millions of chunks, this would take hours per query.

A bi-encoder pays the encoding cost once at indexing time, and approximate nearest-neighbour search makes retrieval roughly constant time even at large scale. So we set the design in two stages: use the cheap bi-encoder to generate a small candidate set from millions of initial chunks with good recall, then spend the expensive cross-encoder only on those few to get the precision and rerank effectively.